Machines & power tools

National 5 · Machines, materials & finishing · Worksheet

WOODWORK WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ use the pillar drill, mortise machine, belt and disc sanders safely
- ☐ use portable power tools safely and make pre-use checks
- ☐ describe machine care and maintenance for my log book

I DO — Worked example

Read how a full-mark answer is built

A pupil uses a belt sander to shape the end of a part.

1 Describe two safety precautions when using the belt sander. (2 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ Keep fingers away from the belt
- ✓ Use the table or fence to support the work

WE DO — Guided practice

Use the starters and word bank to help you

A pupil notices the guard on the pillar drill is damaged.

2 State what the pupil should do. (1 mark)

Sentence starters

1. …

Word bank

teacher · report

Tip: 1 marks = 1 separate, developed points.

YOU DO — On your own

No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which machine cuts square mortises?

☐ Pillar drill ☐ Mortise machine ☐ Disc sander ☐ Jigsaw

b) What is a safety check before using a jigsaw?

☐ Check the cable for damage ☐ Paint the blade ☐ Remove the guard ☐ Wet the wood

c) Which side of a disc sander should you use?

☐ The upward-moving side ☐ The downward-moving side ☐ Either ☐ The edge

d) How should work be held on a pillar drill?

☐ By hand ☐ In a machine vice or clamped ☐ On your knee ☐ It doesn't matter

4 Key terms: write the letter of the matching description next to each term.

1. **Pillar drill safety**	____	**A**	Hollow chisel and auger chop square mortises
2. **Mortise machine**	____	**B**	Cuts curves in boards and sheet material
3. **Disc sander**	____	**C**	Clamp work in a machine vice and remove the chuck key
4. **Jigsaw**	____	**D**	Check cable, plug, guards and blade; clamp the work
5. **Before using a power tool**	____	**E**	Report it and don't use it
6. **Faulty machine**	____	**F**	Use the downward-moving side of the disc

A pupil uses a jigsaw to cut a curve in plywood.

5 Describe two safety checks before using the jigsaw. (2 marks)

Ten identical mortises are needed in table legs.

6 Name a machine that would cut the mortises and give one advantage. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?